WEEK 3 · DAY 3 · OVERVIEW

Exploratory Data Analysis



Created by **Jeff Lenamon**

ABOUT 60 MIN WITH EXERCISES / 8 SECTIONS / 7 EXERCISES

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

The Desk Lead's Note

Ops sent the month-end extract, holdings_messy.csv, as of 2026-09-30. Their cover note says 200 bonds, 489,000,000 of face, 490,819,215 of market value. I need three numbers for the pack: market value, average OAS, average yield. Before any of them leaves the desk, tell me if the file can be trusted. Fix what you can prove, flag what you cannot, show me what each fix does to the three numbers, and give me the list for ops. Then make it run on next month's file in one line.

ABOUT 60 MIN WITH EXERCISES / 8 SECTIONS / 7 EXERCISES

Your Route Through Day 3

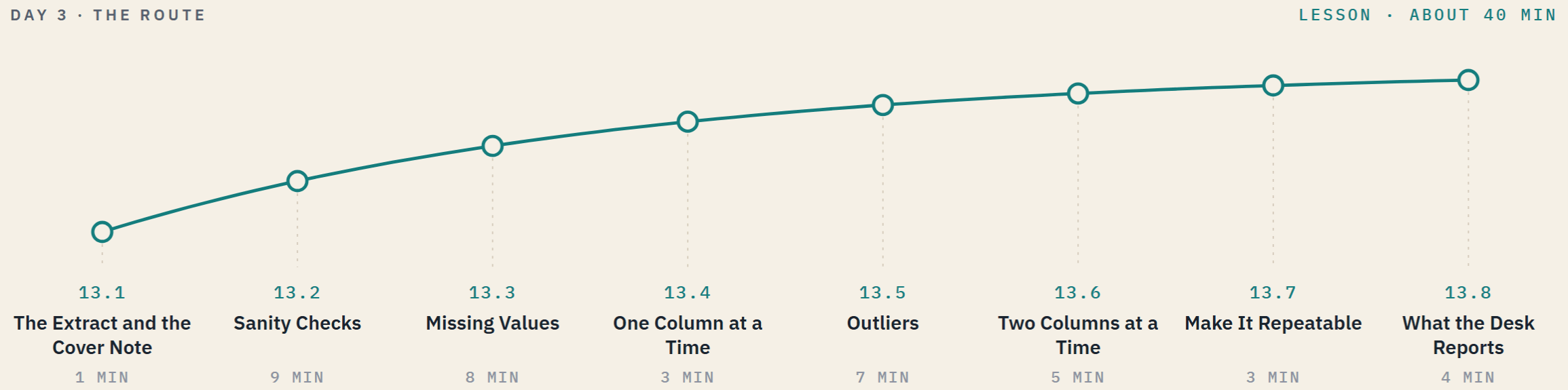
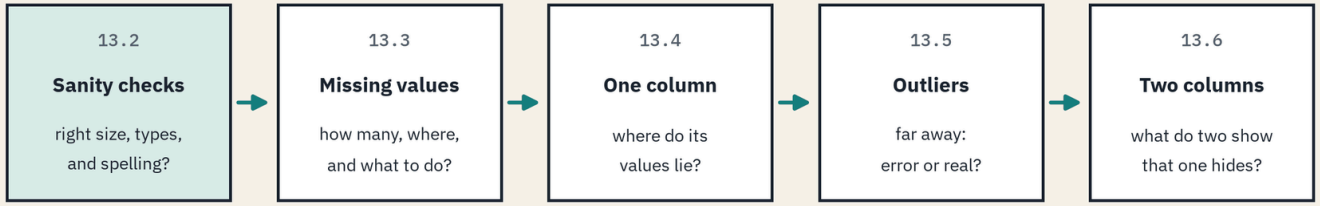


FIGURE 2 Eight sections, in order, with about how long each takes.

Time. About 40 minutes for the lesson, then about 21 for the seven exercises.

13.1 The Extract and the Cover Note

The EDA route: five questions, in this order



Three rules all day:

1. the file as received is never changed: every fix is made on a copy
2. every fix is a decision, stated with its evidence
3. every fix shows its effect on the three numbers

FIGURE 1 The day's five questions in order, under three rules: keep the file as received, log each fix, show its effect.

THE COVER NOTE

```
expected_bonds = 200
expected_face = 489000000
expected_value = 490819215
```

THE THREE NUMBERS AS THE FILE STANDS

```
headline('As received', raw)
```

```
As received | rows: 204 | market value: 497769925.0 | OAS: 187.83 | yield: 6.051
```

204 rows against 200: a finding before the first check.

13.2 Sanity Checks

DATES HELD AS TEXT

```
raw['price_date'].max() - raw['price_date'].min()
```

```
TypeError: unsupported operand type(s) for -: 'str' and 'str'
```

CONVERT IN A WORKING COPY AND COUNT

```
for column in ['as_of_date', 'maturity', 'price_date']:
    book[column] = pd.to_datetime(book[column], format='%Y-%m-%d', errors='coerce')
    print(column, 'not read:', book[column].isna().sum())
```

```
as_of_date not read: 0
maturity not read: 0
price_date not read: 0
```

Every date read; always print that count, because a coerce turns a bad date into a silent gap.

Copies and the Excel Checks

ROWS SENT TWICE

```
copies = book[book.duplicated()]
```

```
Copies at rows: [200, 201, 202, 203]
```

DECISION 1

```
book = book.drop_duplicates().reset_index(drop=True)
```

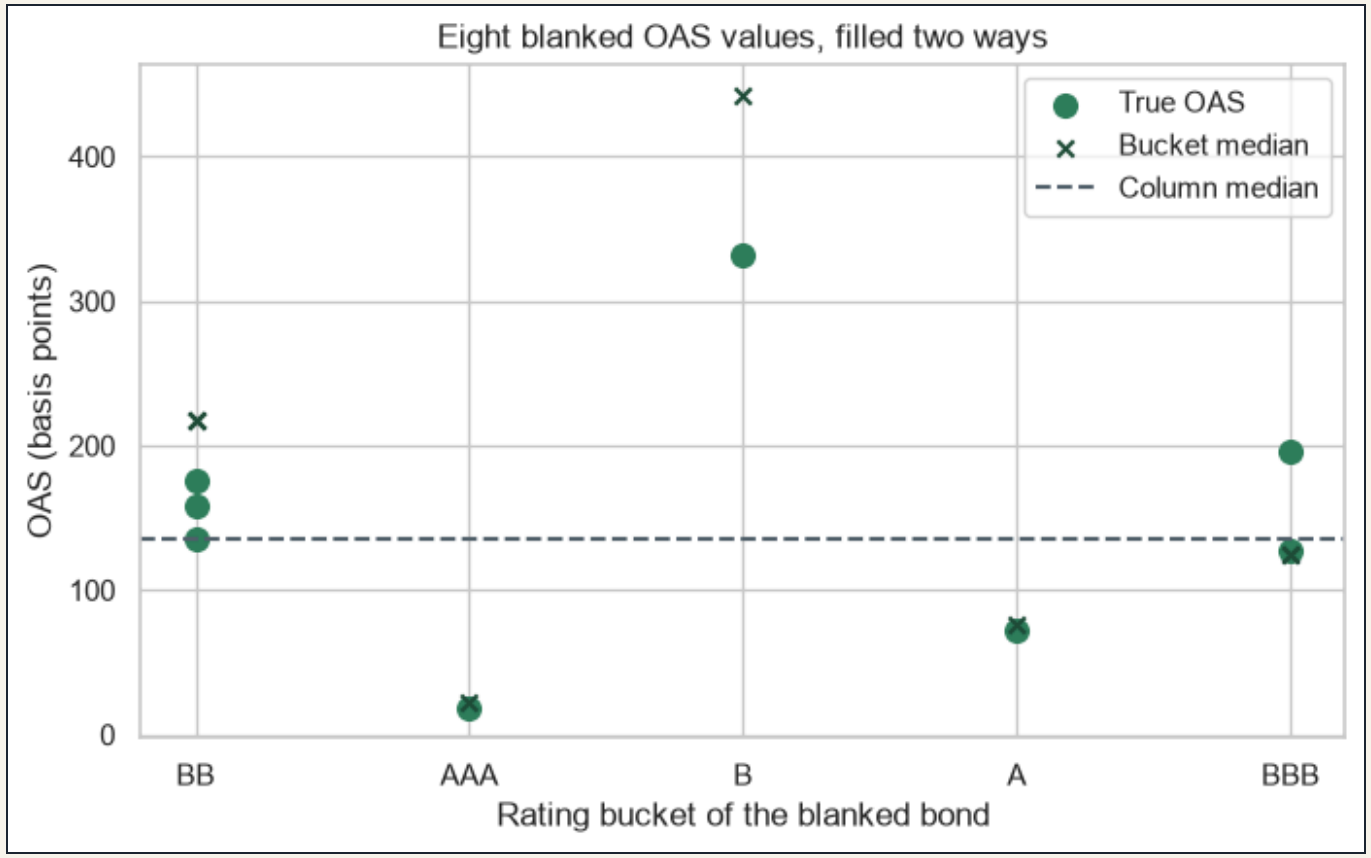
```
Off the note: 0 0 0.0 | raw still has 204 rows
```

EXCEL SIDE BY SIDE

CHECK	PANDAS	EXCEL
Rows and totals	`len(raw)`, `.sum()`	`=COUNTA(B2:B205)`, `=SUM(S2:S205)`
Number or text	`dtypes`	`=ISNUMBER(I2)`, `=ISTEXT(I2)`
Words for missing	`na_values=`	Power Query, Replace Values with `null`
Rows sent twice	`duplicated()`	`=COUNTIF(B:B, B2)>1`; Remove Duplicates deletes in place
Untidy text	`.str.strip().str.upper()`	`=UPPER(TRIM(C2))`
On the scale	`isin()`	`=COUNTIF(ratings!A:A, F2)>0`; Data Validation, List

Two fixes the file proves, one flag it cannot settle. Next, what is not there.

13.3 Missing Values



COUNT THE GAPS

```
print('Gaps:', gaps[gaps > 0].to_dict())
```

```
Gaps: {'rating': 5}
```

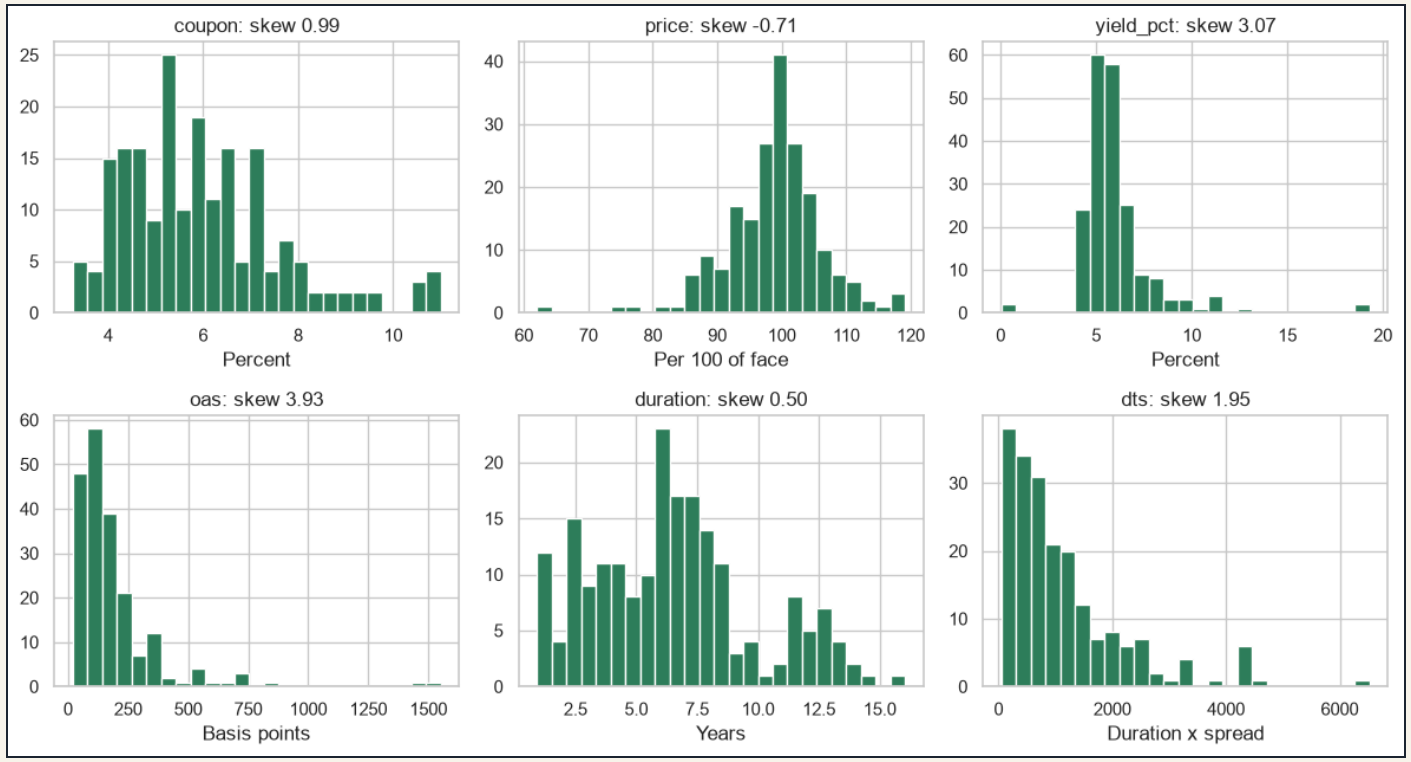
A HUNDRED DRAWS

```
average_miss('oas')
```

```
oas | column median: 102.57 | bucket median: 55.37
```

Text gets an honest label; a number gets its group's median, tested first. Now the values themselves.

13.4 One Column at a Time



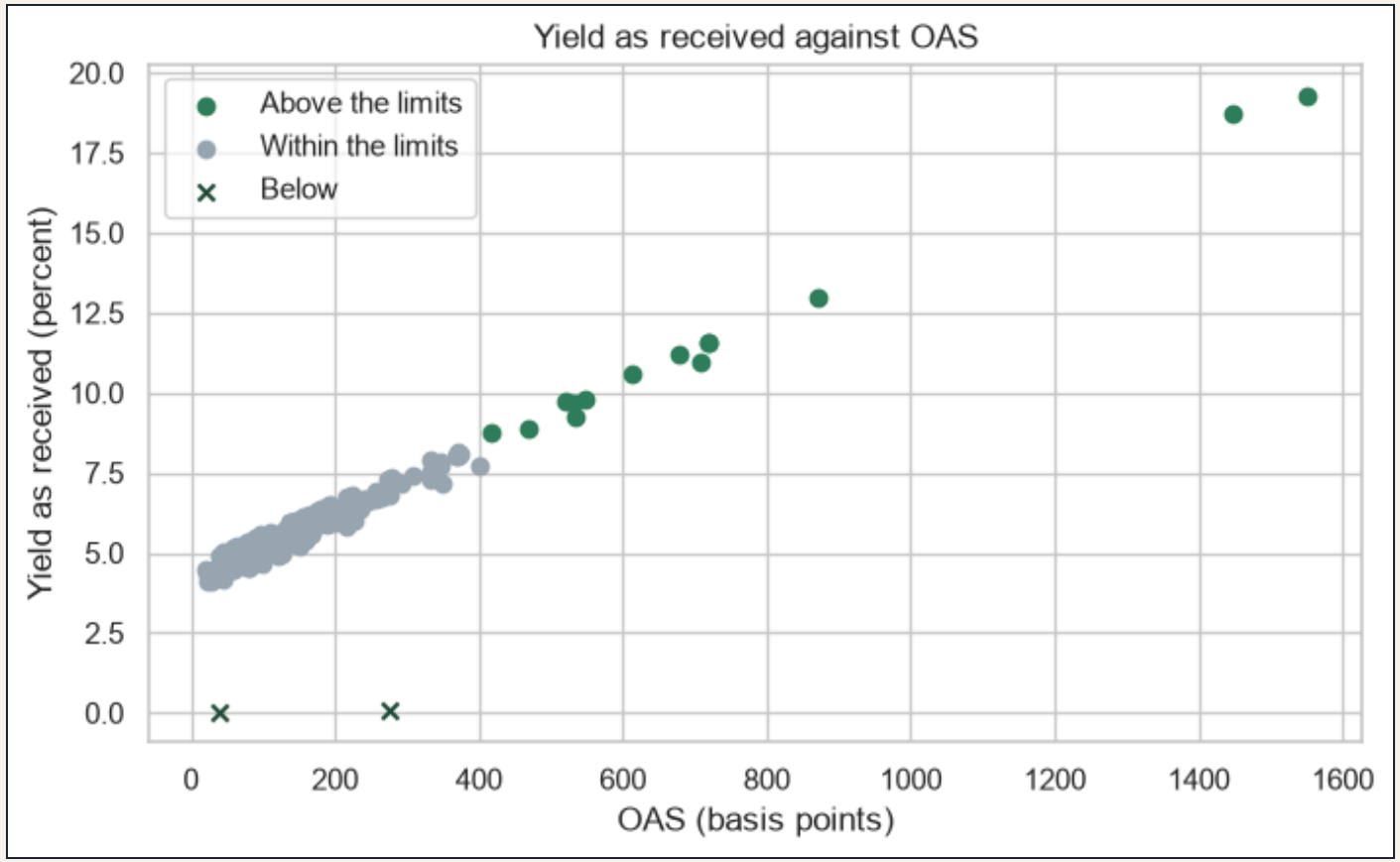
MEAN, MEDIAN, AND SKEW

```
book[columns].agg(['mean', 'median', 'skew']).round(2)
```

	PRICE	YIELD_PCT	OAS
mean	98.83	6.03	185.75
median	99.42	5.53	135.50
skew	-0.71	3.07	3.93

One minimum looks wrong and a few bonds stand far out. Next: error, or real.

13.5 Outliers



THE IQR RULE

```
low_yield = book['yield_pct'] < lower_limit

Below: 2 | above: 14 | inside: 184
```

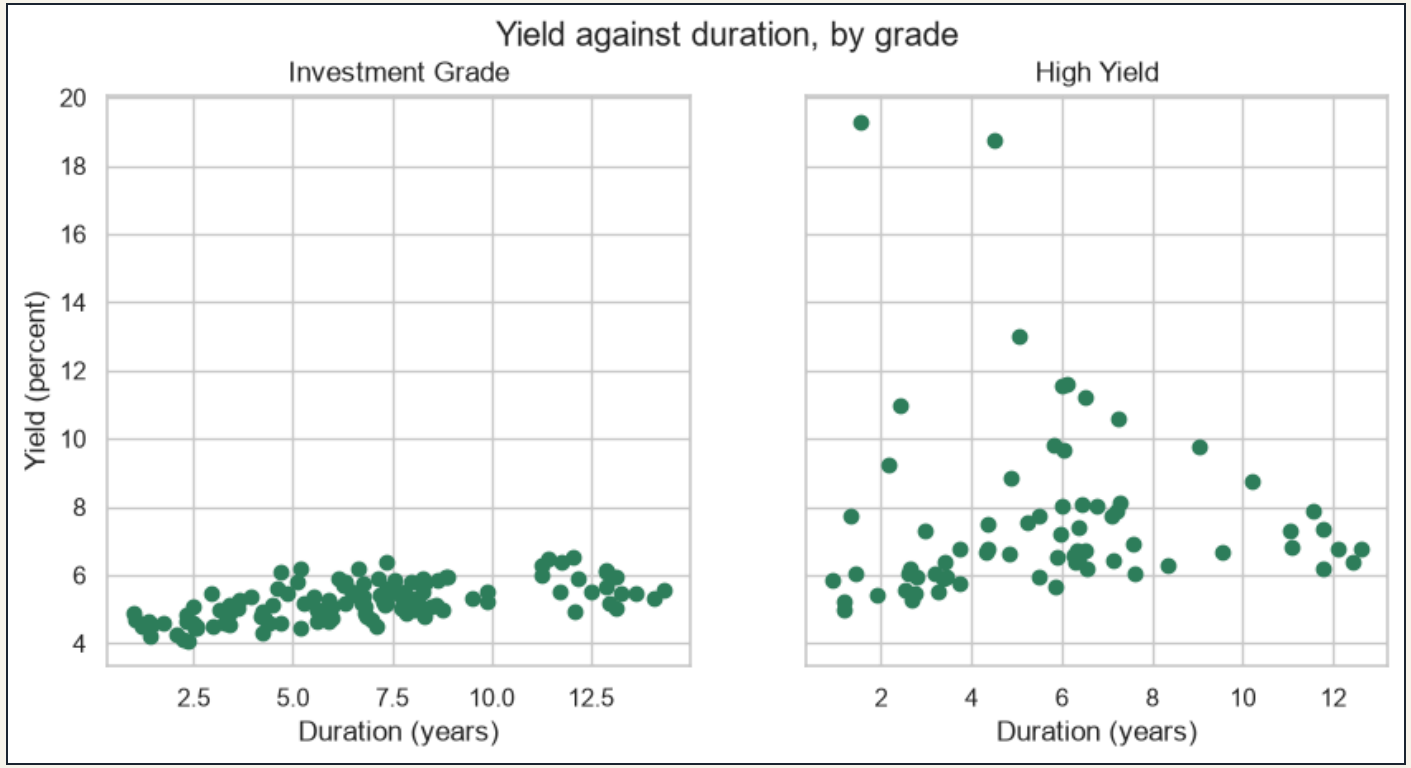
YIELD LESS SPREAD, THE TWO LOWEST

```
base = book['yield_pct'] - book['oas'] / 100

The two: [-0.333, -2.677]
The other 198: 3.691 to 4.585
```

An error breaks a relationship; a real extreme keeps it.

13.6 Two Columns at a Time



ONE NUMBER, TWO GRADES

```
Yield and OAS, as received: 0.95
Yield and OAS, corrected: 0.99
Duration and yield, book: 0.01
Investment Grade 125 bonds: 0.61
High Yield 70 bonds: 0.0
```

DURATION INSIDE MATURITY

```
Duration above years left: 0
```

The file is understood. Now the day in one place.

13.7 Make It Repeatable

THE DAY IN ONE FUNCTION

```
def clean_holdings(raw, ratings):
    # decision 1: remove the rows sent twice
    # decision 2: tidy the tickers
    return book
```

ONE LINE

```
rebuilt = clean_holdings(raw, ratings)
```

```
(200, 22) | same as today's book: True | raw: (204, 19)
```

CHECKS THAT STOP

```
All three control totals match the cover note.
```

THE DECISION LOG

	PROBLEM	ROWS	DECISION
0	Rows sent twice	4	removed
1	Untidy tickers	8	tidied
2	Stale prices	6	flagged for ops
3	No rating	5	labelled Not rated
4	Yields as decimals	2	times 100
5	Very wide OAS	14	kept

THE SAME CHECK ON THE FILE AS RECEIVED

```
AssertionError: 204 rows; the note says 200
```

Next month is one line, and the log travels with the table.

13.8 What the Desk Reports

The three numbers, decision by decision

	market value	mean OAS, bp	mean yield, %
As received	497,769,925	187.83	6.051
Decision 1: remove 4 copies	490,819,215	185.75	6.028
Decisions 2, 3, 4	unchanged	unchanged	unchanged
Decision 5: 2 yields x 100	unchanged	unchanged	6.088
Decision 6: keep the 14	unchanged	unchanged	unchanged
As corrected	490,819,215	185.75	6.088

green: the numbers a decision moved, then the three for the pack

FIGURE 5 Market value, OAS, and yield after each decision, from the file as received to the file as corrected.

THE FIVE MISSING RATINGS, FOUND

	TICKER	RATING	GRADE
0	CORL	A-	Investment Grade
1	GALX	B-	High Yield
2	MORX	BB-	High Yield
3	NEVE	A-	Investment Grade
4	SYLX	A+	Investment Grade

WHERE THE TWO FILES DIFFER

```
Same CUSIPs, same order: True
rating 5
price_date 6
```

Pack, list for ops, workbook, and next month in one line: the note is answered.

You Can Now

- Check a file against its control totals before using it.
- Find dates and numbers held as text, convert them, and count what failed.
- Find repeated rows, untidy text, and values off a list.
- Leave, drop, guess, or label a gap, and say what each does to a total.
- Flag outliers with the IQR rule and z-scores, and tell an error from a real extreme.
- Package the day as a function with checks, a decision log, and a workbook.

NEXT

WEEK 3 · DAY 4

Case Study: Treasury Yields

Daily Treasury par yields from 2015 to 2026 are loaded as a time series and described by tenor, by year, and day by day.